PROJECT PRESENTATION • ACADEMIC YEAR 2026–2027

# NETVISION

## High-Dimensional Network Log Visualisation Using PCA and t-SNE

**Usharani | 2520090049**
**Anushka | 2520090168**

Department: Computer Science & Information Technology
College/University: KLH / KL University
Guide/Supervisor: [Add guide name]

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# Introduction

- Network logs contain many features that are difficult to inspect directly.
- High-dimensional data can hide patterns and make visual interpretation challenging.
- NETVISION combines dimensionality reduction, visualization and anomaly detection.
- The project uses the UNSW-NB15 network intrusion dataset.

# Problem Statement

- Large network datasets contain many attributes and complex relationships.
- Inspecting records one by one is slow and makes unusual patterns harder to notice.
- Raw tables do not provide an intuitive view of high-dimensional structure.
- Analysts and learners need clear visual summaries to explore network records.

# Motivation

- Make complex network-log data easier to explore and interpret.
- Use visual patterns to compare normal and attack-labelled traffic.
- Explore unsupervised methods that can flag unusual records without attack labels during training.
- Create a reusable dashboard for network-data exploration.

# Objectives

### 01 • Explore

Inspect data structure, data types, missing values and duplicates.

### 02 • Reduce

Apply PCA to reduce encoded features to 30 components.

### 03 • Visualize

Use t-SNE on a 2,000-record sample.

### 04 • Detect

Use Isolation Forest to flag potential anomalies.

### 05 • Present

Display plots, summaries and exported results.

# Existing System

## Current approach

- Inspect raw tables and summary statistics
- Use separate scripts for plots and anomaly detection
- Manually compare normal and attack-labelled records

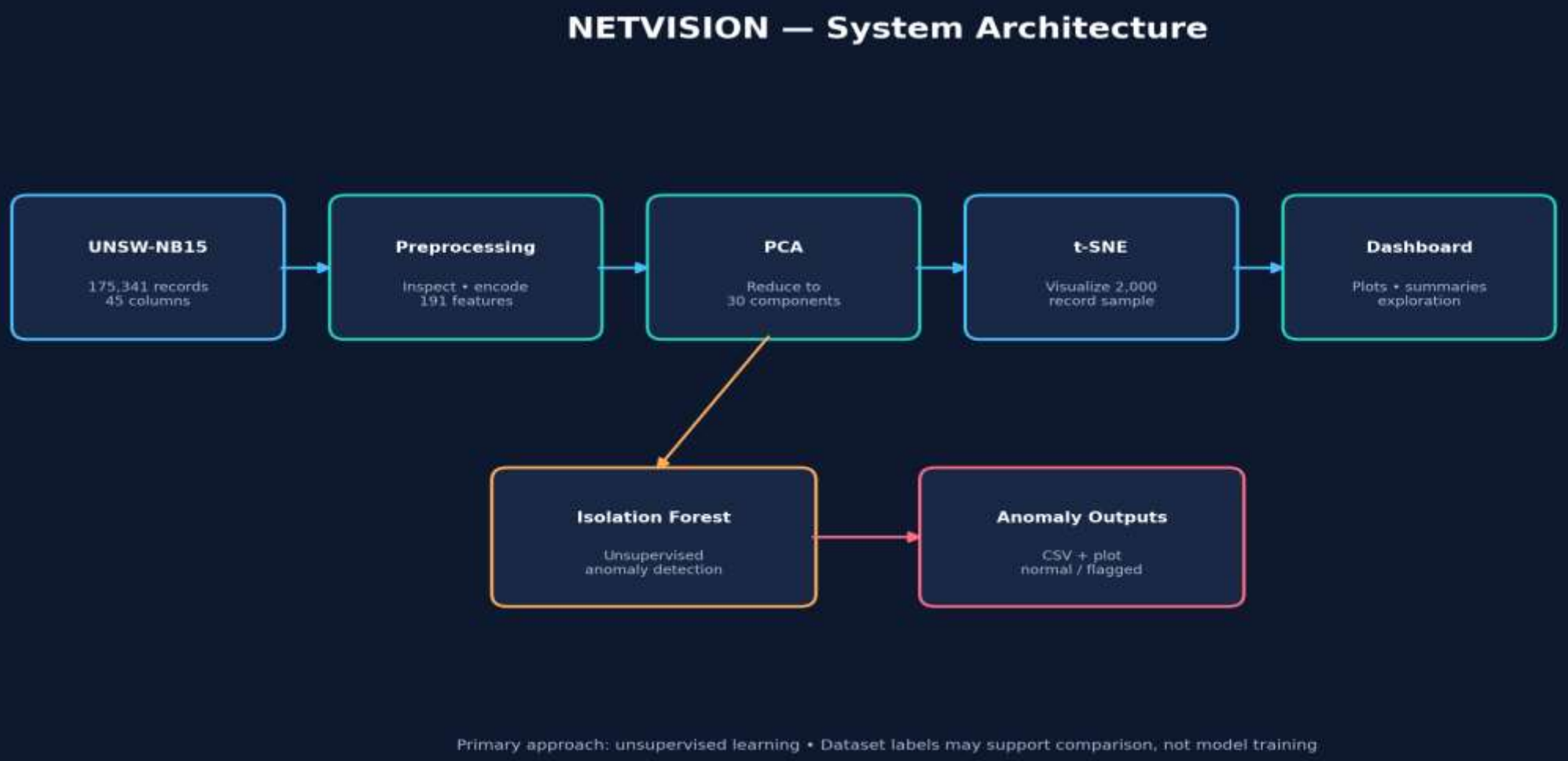
## Limitations

- High feature count is difficult to interpret
- Manual exploration can be time-consuming
- Outputs may be scattered across files
- Anomaly flags alone do not confirm attacks

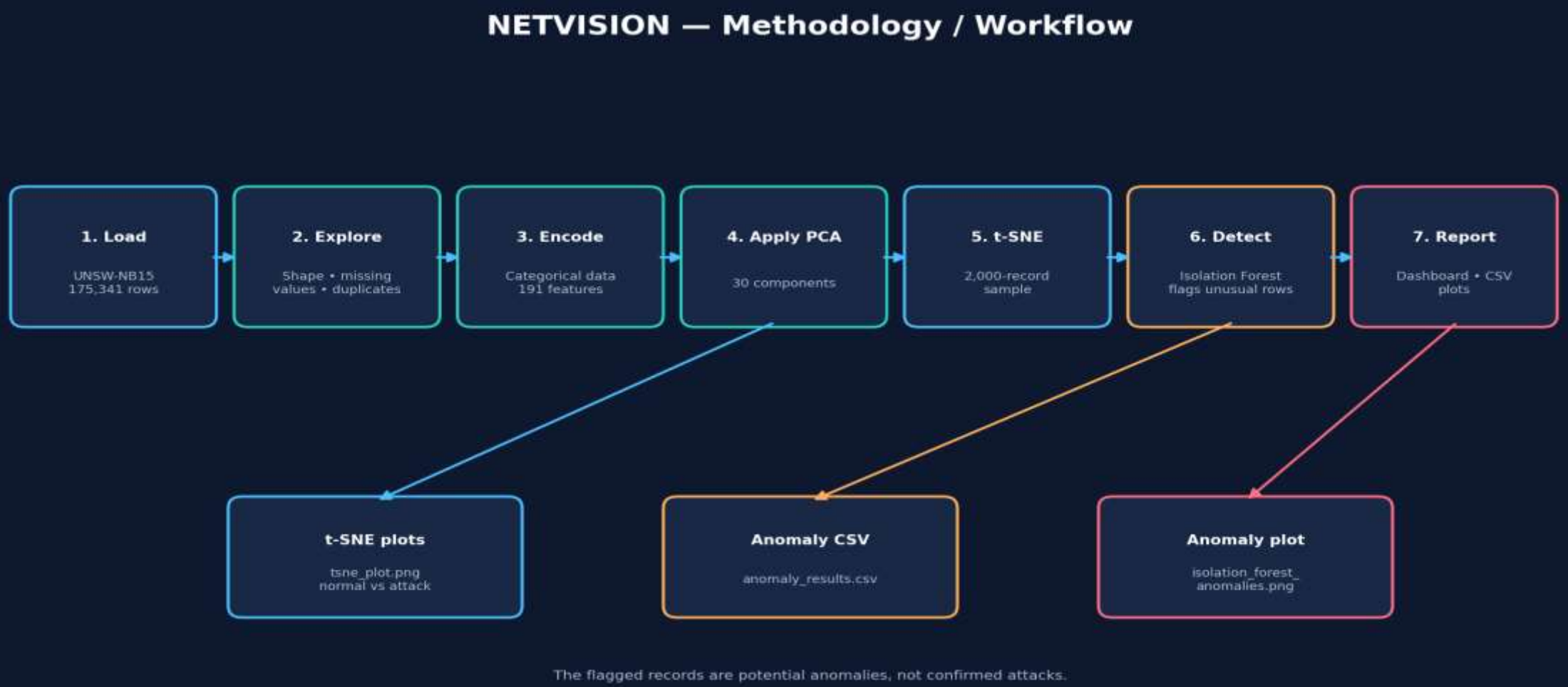
# Proposed System

- A unified workflow for data inspection, encoding, PCA, t-SNE and anomaly detection.
- A 2D visual representation helps explore high-dimensional patterns.
- Isolation Forest produces a separate list of records flagged as unusual.
- Dashboard and exports support exploration; flagged records require further investigation.

# System Architecture / Block Diagram



# Methodology / Workflow



# Technology Stack

**Language**

Python

**Dashboard**

Streamlit

**Data handling**

Pandas • NumPy

**ML library**

Scikit-learn

**Visualization**

Matplotlib / Plotly

Database, hardware and cloud services are not required for the current offline dataset workflow.

# System Modules / Components

### Data Ingestion

Load UNSW-NB15 and inspect its shape.

### Preprocessing

Check data quality and encode categories.

### PCA + t-SNE

Reduce dimensions and create visual plots.

### Anomaly Detection

Isolation Forest flags unusual records.

### Dashboard / Export

Display summaries and save CSV / plots.

# Implementation

## Processing steps

- Load dataset and inspect data quality
- Encode categorical columns
- Run PCA, t-SNE and Isolation Forest

## Generated outputs

- t-SNE plots
- anomaly_results.csv
- isolation_forest_anomalies.png

## Recorded matrix sizes

- Encoded data: 175,341 × 191
- PCA data: 175,341 × 30

## Dashboard

Streamlit is used to explore summaries and results.

Add your current dashboard screenshots before presenting, if required.

# Dataset / Input Data

**Dataset**

UNSW-NB15

**Raw dimensions**

175,341 records
45 columns

**Data quality**

0 missing values
0 duplicates reported

**After encoding**

191 features

**PCA output**

30 components

Values reflect the recorded project run; verify against the exact dataset file used.

# Algorithm / Model

### PCA

Unsupervised dimensionality reduction; creates a compact representation.

### t-SNE

Unsupervised embedding to visualize local structure in 2D.

### Isolation Forest

Unsupervised anomaly detection; flags unusual records.

**Overall: primarily UNSUPERVISED learning. Dataset labels may color or compare plots, but do not make these algorithms supervised.**

**Anomaly flags indicate unusual records, not confirmed malicious activity.**

# Results

**Normal records**

166,574

**Flagged anomalies**

8,767

**Flagged proportion**

5.00%

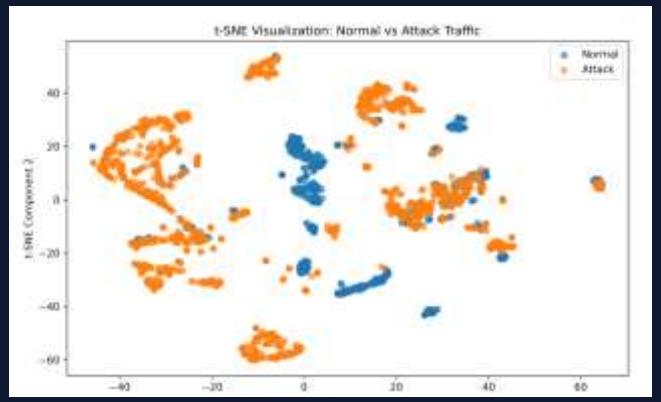
- PCA reduced 191 encoded features to 30 components.
- t-SNE visualization used a sample of 2,000 records.
- No accuracy, precision, recall or F1-score is claimed because this pipeline is not a supervised classifier.

# Screenshots / Demo

## Dashboard overview




## t-SNE visualization




## Anomaly results




These are clearly marked placeholders; replace with screenshots from your actual running project.

# Advantages

- Makes high-dimensional data easier to inspect.
- Combines visualization and anomaly outputs in one workflow.
- Uses unsupervised methods that do not require attack labels for training.
- Exports plots and records for further analysis.
- Supports exploratory review, rather than automatically confirming attacks.

# Applications

- Network-security education and research.
- Initial exploration of large network-traffic datasets.
- Supporting analyst triage by highlighting unusual records.
- Visual comparison of normal and attack-labelled traffic patterns.
- Demonstrations of PCA, t-SNE and unsupervised anomaly detection.

# Limitations

- t-SNE visualization uses a sample of 2,000 records, not the full dataset.
- Visual clusters do not prove that classes are cleanly separated.
- Isolation Forest flags may include benign unusual activity.
- Results depend on preprocessing, selected features and model parameters.
- A flagged record needs validation before being treated as an attack.

# Future Scope

- Add interactive filters and record-level drill-downs.
- Compare anomaly flags with ground-truth labels using suitable evaluation metrics.
- Compare Isolation Forest with other anomaly-detection algorithms.
- Improve sampling and processing for larger datasets.
- Deploy the dashboard for controlled use by analysts or researchers.

# Conclusion

- NETVISION provides a workflow for exploring high-dimensional network logs.
- PCA and t-SNE support compact representation and visual inspection.
- Isolation Forest flagged 8,767 records as potential anomalies in the recorded run.
- The project is primarily unsupervised and supports investigation rather than definitive attack decisions.

**Thank you • Questions?**